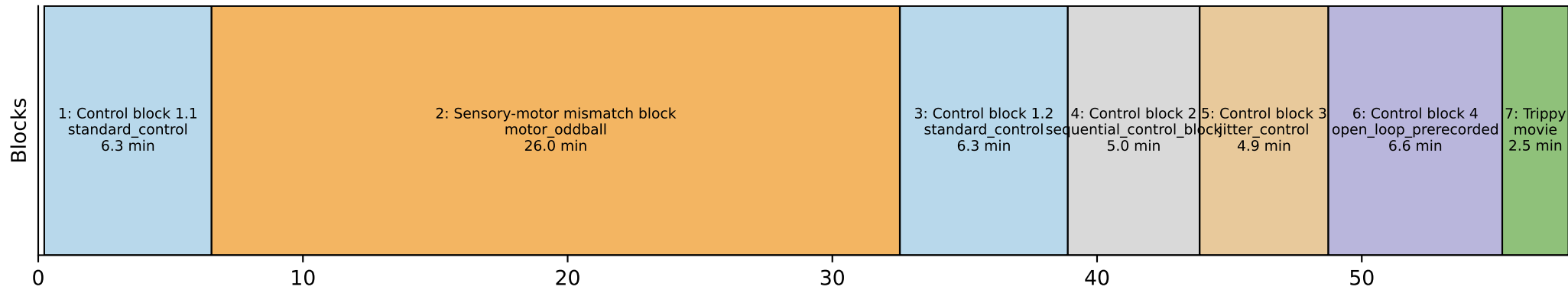
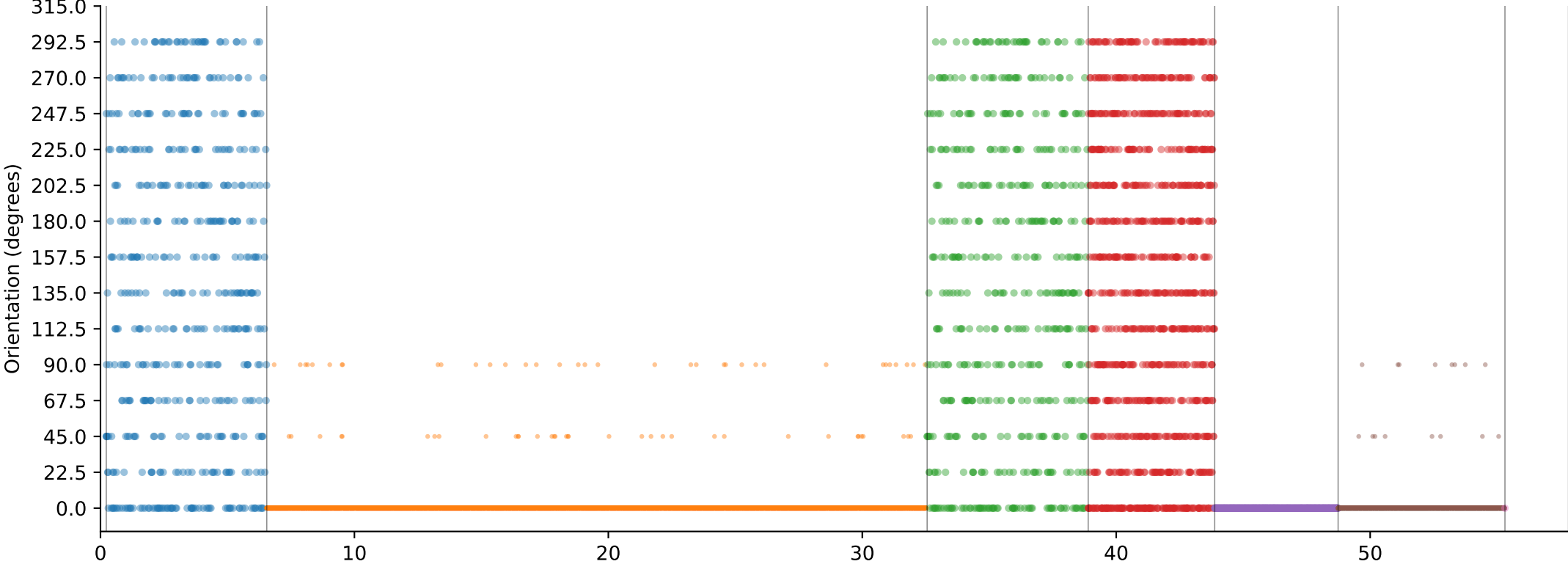


VW01 2026-05-22 Sensory-Motor Paradigm Structure

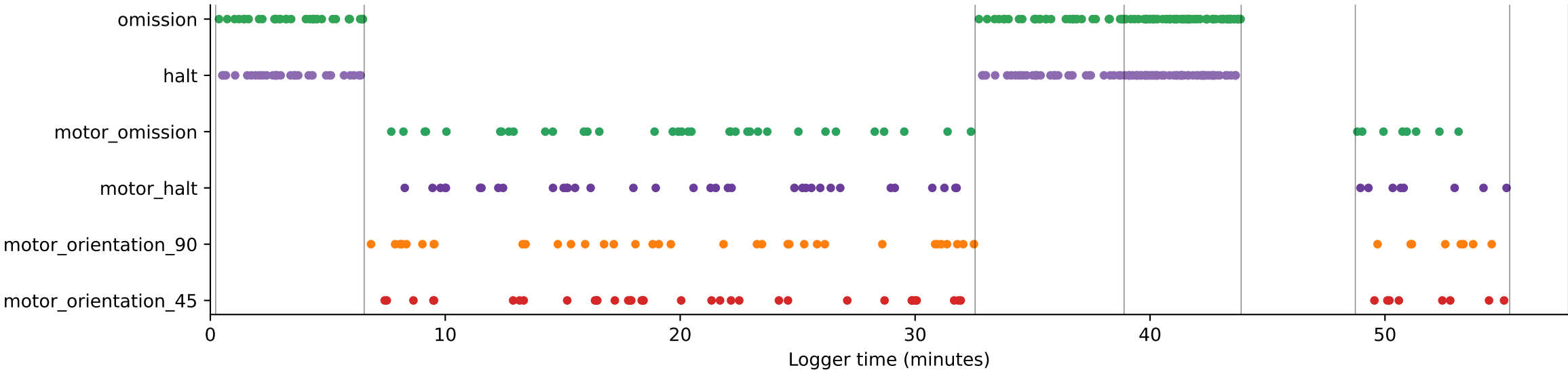
Recorded Block Structure



Performed Grating Orientations Over Time



Recorded Halt, Omission, And Orientation-Oddball Events



GitHub Example Versus Recorded Performed-Stimulus Verification

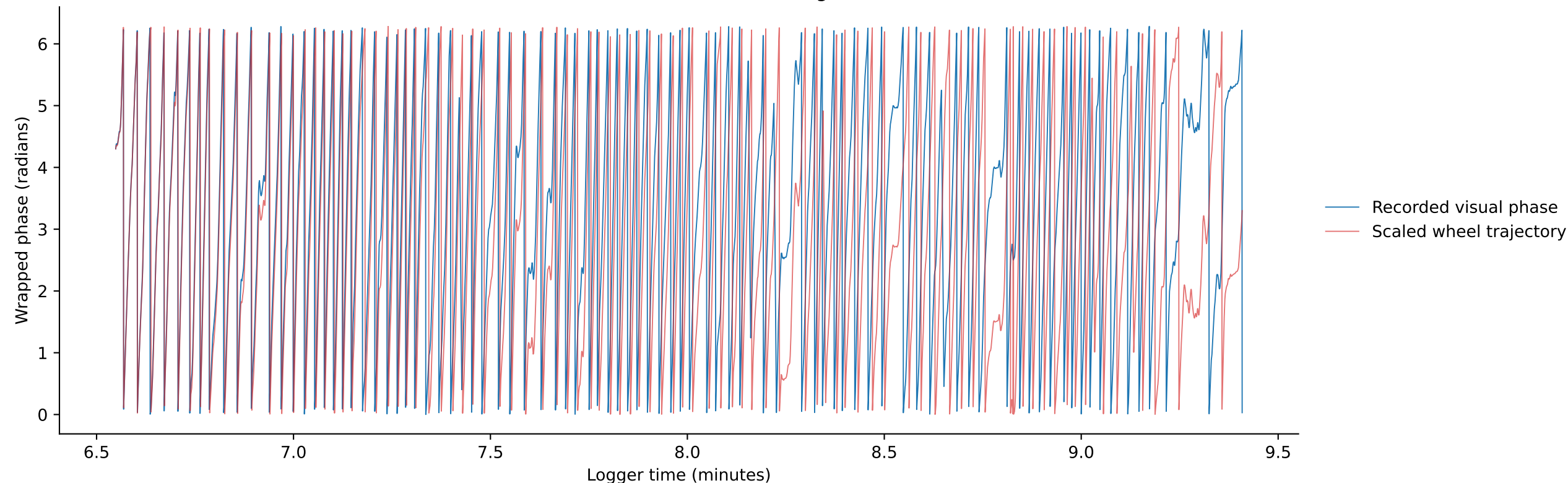
Block	Label	Expected	Recorded	Types	Timing	Semantics	Note
1	Control block 1.1	544	544	PASS	PASS	PASS	all semantic fields and row order match
2	Sensory-motor mismatch block	45461	45461	PASS	PASS	PASS	planned wheel phase replaced by recorded numeric phase
3	Control block 1.2	544	544	PASS	PASS	PASS	all semantic fields and row order match
4	Control block 2	1120	1120	PASS	PASS	PASS	all semantic fields and row order match
5	Control block 3	556	556	PASS	PASS	PASS	all semantic fields and row order match
6	Control block 4	11520	11520	PASS	PASS	PASS	all semantic fields and row order match
7	Trippy	2	1	FAIL	PASS	FAIL	row count differs (2 expected, 1 recorded)

- Blocks 1-6 match the official Mindscope sensorimotor_mismatch example in row count, condition counts, semantic values, and order.
- Block 2 is expected to differ only in Phase: the planned value 'wheel' is replaced online by the numeric wheel-derived phase.
- The GitHub example has two Trippy rows; the recording has one and its logger StimEnd is missing.
- The GitHub example then contains Block 8 Zebra, Block 9 RF mapping; these blocks are absent from the recorded output.

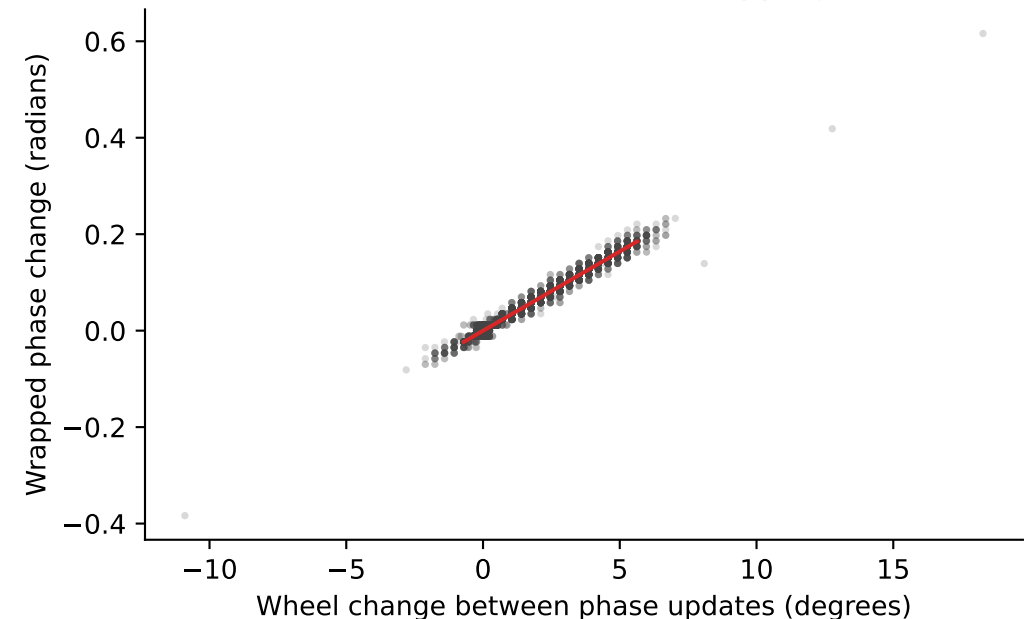
PASS verifies Bonsai commands/logger timing, not physical monitor output. A usable hardware photodiode is required for independent display validation.

Block 2 Wheel-to-Visual-Phase Verification

Block 2 Wheel And Standard-Grating Phase



Local Wheel-to-Phase Mapping



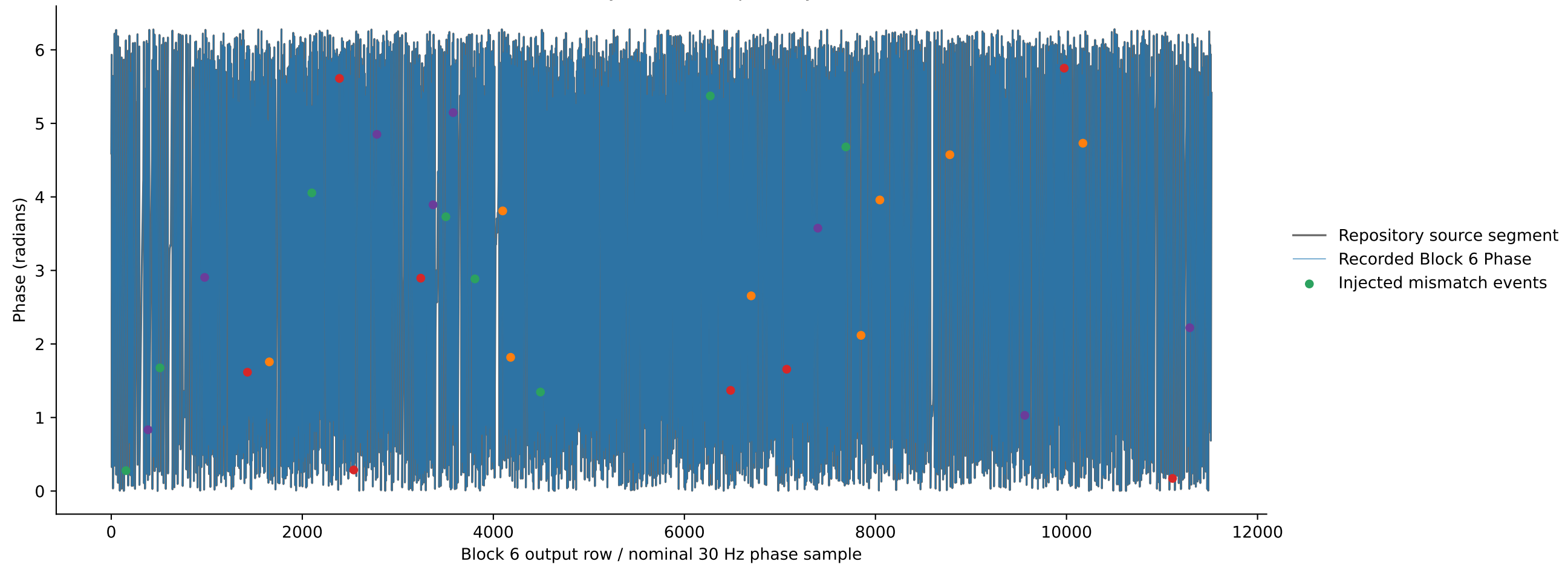
Block 2 standard updates: 45,321
Wheel logger samples: 88,828
Valid local update pairs: 45,234

Correlation: 0.9917
R-squared: 0.9835
Fitted gain: 0.032887 rad phase / wheel degree
1.8843 deg phase / wheel degree
Median absolute residual: 0.000443 rad
95th percentile residual: 0.012523 rad
Residual RMSE: 0.006923 rad

PASS criterion used here: R-squared ≥ 0.95 and correlation ≥ 0.95 .
Result: PASS

The gain is estimated from this recording. The workflow externalizes wheel/screen coupling parameters, so the configured gain cannot be recovered from the CSV files alone.

Block 6 Recorded Phase Exactly Matches Repository Prerecorded Source



Source file:
running_phases/250815165708_366122_3dd8df03-615d-4633-a5e3-a6bb3635c3b7_running_phase_wheel.csv

Unique matching source interval: zero-based rows 38533:50053
Source Timestamp values: 867.764859 to 1126.230602 s
Rows compared: 11,520; maximum absolute phase error: 8.88e-16

Generation logic: select a contiguous 6.4-minute nominal 30 Hz segment using deterministic seed variant=0,
then replace 8 rows of each mismatch type while preserving the source phase values.
Mismatch counts: motor_halt=8, motor_omission=8, motor_orientation_45=8, motor_orientation_90=8

Conclusion: Block 6 replays a phase trajectory extracted from an earlier repository wheel recording;
it is not a replay of VW01's May 22 Block 2 wheel trajectory.